

Frank Goortani, TOGAF, PMP

Senior AI/ML Engineer • Production LLM Systems • Contract/Freelance

Senior AI/ML Engineer with **25+ years** full-stack experience and **deep production LLM expertise**. Built **award-winning ELLE AI Decision Engine** at Uber using LangChain, RAG systems, and AI agents, processing millions of decisions across global platform. Expert in **Python** (FastAPI, LangChain, LangGraph), **Go** (microservices, YARPC), and **React/TypeScript** (enterprise web apps), with proven ability to ship production LLM features fast.

Available for contract work (10-40 hrs/week) where I can apply production AI/ML expertise to help companies ship LLM-powered features faster.

Tools, Frameworks, Libraries, Languages, Databases

AI/ML & LLM Frameworks:

LangChain, LangGraph, FastMCP, MCP (Model Context Protocol), CrewAI, Ollama, Prompt Engineering, Agentic Workflows, RAG Systems

Agentic Coding & AI Tools:

Claude Code, Cursor, GitHub Copilot, Aider, Codex, AI-Assisted Development, MCP Development, Agentic Coding Workflows

LLM Platforms:

OpenAI (GPT-4, GPT-3.5), Anthropic Claude (Sonnet, Opus), Google Gemini, Cohere, Meta LLMs, Hugging Face Models, Open Source LLMs

Vector Databases & Embeddings:

OpenSearch, Pinecone, Qdrant, Chroma, Weaviate, Embedding Models, Semantic Search, RAG (Retrieval Augmented Generation)

Backend:

Python (FastAPI, LangFx, Pydantic, Uvicorn, async/await), Go (YARPC, gRPC, microservices), Node.js, Java (Spring Boot), C#

API & Integration:

GraphQL, gRPC, REST, Protocol Buffers, Apache Thrift, Swagger, OAuth2/OIDC, JWT

Frontend:

React 18, TypeScript 5.7, Fusion.js, NextJS, TanStack React Query, Redux, Angular, VueJs, BaseUI, shadcn/ui, Tailwind CSS, Material Design

Cloud Platforms:

Google Cloud (GCP Cloud Run, Firestore, Firebase), AWS (EC2, Lambda, S3, RDS), Azure (VMs, Functions, CosmosDB)

Container & Orchestration:

Docker, Kubernetes (K8S), Helm, Istio, Rancher, Google Kubernetes Engine

Data & Messaging:

Apache Kafka, Cadence, Temporal, Apache Spark (PySpark), Presto, Hive, HDFS, Airflow-like workflows

Databases:

Cassandra, MySQL, PostgreSQL, MongoDB, Redis, DynamoDB, Elasticsearch, CouchDB, SQL Server

DevOps & CI/CD:

Bazel, GitHub Actions, Jenkins, Bamboo, Circle CI, Terraform, Ansible, Chef, Puppet

Monitoring & Observability:

M3, Prometheus, Grafana, Datadog, Splunk, Zap Logging, Jaeger (distributed tracing), Kibana

Testing:

Jest, Pytest, React Testing Library, Playwright, Selenium, Mockito, Unit Testing, E2E Testing

Security & Identity:

Charter (IAM), SPIFFE/SPIRE, RBAC, GDPR/CCPA Compliance, OAuth2, OpenID, OWASP, Data Privacy

Languages:

Python, Go (Golang), JavaScript, TypeScript, Java, C#, Ruby, Swift, Kotlin, SQL, HTML5, CSS3, YAML, JSON

Skills and Achievements

- **Production LLM Systems:** Multi-agent architectures, RAG pipelines, prompt engineering,

agentic workflows

- **Agentic Coding & AI-Assisted Development:** Expert in Claude Code, Cursor, GitHub Copilot - achieving 3-5x productivity through AI-powered development workflows, MCP (Model Context Protocol) development, agentic coding patterns
- **Distributed Systems:** Microservices, API platforms, workflow orchestration, event-driven architecture
- **Full-Stack Development:** Python backend, React/TypeScript frontend, Go microservices, cloud-native applications
- **AI/ML Engineering:** LangChain/LangGraph, vector databases, semantic search, model fine-tuning, ML pipelines
- **Data Engineering:** Apache Spark, Kafka streaming, ETL/ELT pipelines, real-time analytics, big data processing
- **Cloud Architecture:** GCP, AWS, Azure - serverless, containerized, and managed services
- **DevOps & Automation:** CI/CD pipelines, infrastructure as code, container orchestration, monitoring/observability
- **Security Engineering:** Authorization systems, privacy automation, GDPR/CCPA compliance, secure coding practices
- **Technical Leadership:** Architecting complex systems, code reviews, technical mentorship, best practices enforcement
- **Rapid Prototyping:** MVP development, 0→1 builds, leveraging agentic coding tools for accelerated development cycles

Work Experience

Solution Architect

Uber

Jan 2021 - Present

Remote

ELLE (EngsecLLM Engine) - AI Decision Engine

Achievement: Won multiple internal Uber innovation awards for automating Technical Privacy review workflows, reducing manual review time by 70%.

Technical Leadership:

- Architected and built production LLM system processing **millions of security/privacy decisions** across Uber's global platform
- Designed **multi-agent architecture** using LangChain and LangGraph with ReAct and self-reflection patterns
- Implemented **6+ automated workflows:** Technical Privacy Review, Security Guidance, Legal Review, DSAR Automation, Data Classification, Compliance Checking
- Built **RAG system** with OpenSearch vector store for semantic search across security policies, privacy regulations, and compliance documentation
- Developed **authorization middleware** (Go, YARPC, Charter) with **96% test coverage** and dual inbound/outbound protection
- Integrated with **30+ Uber platform services:** USSO (auth), Charter (IAM), Cadence (workflows), M3 (metrics), Zap (logging)

Technologies: Python (FastAPI, LangChain, LangGraph, FastMCP), Go (YARPC, gRPC), OpenAI API, OpenSearch, Cadence workflows, Cassandra, Redis, Kafka, Claude Code (AI-assisted development)

Technical Highlights:

- Implemented prompt engineering strategies achieving 92% accuracy on privacy classification tasks
- Designed agentic workflow orchestration handling 100K+ requests/month with sub-200ms p95 latency
- Built automated monitoring with Zap logging, Tally metrics, and Jaeger distributed tracing
- Leveraged Claude Code and agentic coding tools achieving 3-5x development velocity on AI features

9 Enterprise Privacy Web Applications - Full-Stack Expertise

Led full-stack development of 9+ mission-critical privacy web applications serving Uber's global compliance teams:

- **Elle Web** - EngsecLLM Engine Frontend (React 18, TypeScript, Fusion.js)
- **IRIS Web** - Privacy Web Frontend (GDPR/CCPA compliance automation)
- **DSAR Web** - Subject Access Request Automation (automated workflows reducing manual work by 60%)
- **UDE (User Data Explorer)** - Data discovery and lineage tracking
- **Consents Manager Web** - Contextual consent management (GDPR/CCPA compliant)
- **Web CMP** - IAB TCF v2 Compliant Consent Management Platform (cookie consent across 70+ countries)
- **TPRM QC Copilot Web** - Third-Party Risk Management quality control
- **Agentic AI Observability Web** - Real-time monitoring for AI agent systems
- **Design Review Web** - Privacy design review workflows

Technologies: React 18, TypeScript 5.7, Fusion.js (universal rendering), TanStack React Query, Redux, BaseUI, Styletron (CSS-in-JS), Protocol Buffers (RPC), Jest (34+ test suites), Playwright (E2E testing), Bazel (monorepo builds)

Technical Achievements:

- Implemented **universal rendering** with Fusion.js achieving 40% faster page load times
- Built **type-safe RPC communication** via Protocol Buffers for 30+ backend services
- Comprehensive testing achieving **96% code coverage** across all applications
- Integrated **IAB TCF v2 compliance** (Web CMP) for cookie consent across Uber's global web properties

50+ Security Infrastructure Projects - Python Monorepo

Led development of 50+ security infrastructure projects in Python monorepo spanning AI/ML, data engineering, and automation:

AI/ML Projects:

- LLM-powered security automation using LangChain, LangGraph, and FastMCP (Model Context Protocol)
- Built custom MCP servers for agentic workflows and tool integration (Model Context Protocol development)
- RAG systems for security policy search (OpenSearch vector store, semantic embeddings)
- Multi-agent systems with self-reflection and ReAct patterns
- Container image remediation assistant (LangFx service with JIRA integration)
- Developed AI-assisted coding workflows using Claude Code, Cursor, and GitHub Copilot

Data Engineering:

- Apache Spark pipelines for security event processing
- Kafka streaming for real-time security alerts
- Presto SQL queries for compliance reporting

Technologies: Python 3.9/3.11, LangChain, LangGraph, FastMCP, Apache Spark, Kafka, Presto, OpenSearch, Pinecone, Qdrant, Bazel, pytest

40+ Engineering Security Projects - Go Microservices

Contributed to 40+ Engineering Security projects in Uber's Go monorepo, including core platform services:

Major Contributions:

- **Charter** - Uber's authorization platform (IAM equivalent), designed RBAC policies, integrated Decision Service
- **SPIRE/SPIFFE** - Workload identity framework, implemented service-to-service authentication
- **USSO (Uber SSO)** - Single Sign-On integration across all internal tools
- **Authorization Middleware** - Production-ready middleware with 96% test coverage, dual inbound/outbound protection

- **Distributed Systems** - Go microservices using YARPC, gRPC, Cadence workflows, Kafka messaging

Technologies: Go (Golang), YARPC, gRPC, Protocol Buffers, Apache Thrift, Charter, SPIFFE/SPIRE, authfx, OAuth2, JWT, Cadence, Kafka, Cassandra, MySQL, Redis, Bazel, Docker, Kubernetes

Technical Achievements:

- Solved complex **circular dependency** challenges using MCP Gateway pattern with fx.Lifecycle hooks
- Implemented dual middleware architecture achieving comprehensive endpoint coverage
- Built distributed tracing integration with Jaeger for debugging microservices

AI Solutions Lead

BayRockLabs / VisionZLab

Jan 2021 - Present

Remote

Consulting Engagements:

- Led GenAI implementations for clients across **legal tech, fintech, healthcare, ecommerce, and manufacturing**
- Built production LLM systems using LangChain, RAG architectures, and AI agents
- Implemented full-stack solutions (React, NextJS, Python, Firebase, GCP)
- Delivered technical interviews for top-tier companies (Robinhood, Netflix, Airbnb, Peloton, Gap)

Key Projects:

- **FasterOutcomes** (Legal AI): RAG system for legal document analysis, contract review, compliance checking
- **Counta AI** (Accounting): Multi-agent accounting automation workflows
- **Ecommerce AI**: Product recommendations, inventory optimization, customer behavior prediction
- **Manufacturing AI**: Predictive maintenance, quality control automation

Technologies: Python (LangChain, LangGraph, FastAPI), React, NextJS, Firebase/Firestore, GCP, OpenAI, Anthropic Claude, Pinecone, Qdrant

Industries: Legal Tech • Fintech • Healthcare AI • Ecommerce • Manufacturing • Hospitality

Lead Solution Architect

Home Depot Canada

Dec 2017 - Dec 2019

Toronto, Canada

Team Leadership:

- Led team of **10+ onshore and 50+ offshore developers** for homedepot.ca website
- Architected ecommerce platform migration to microservices architecture
- Built onshore mobile app team from scratch (iOS/Android native development)

Technical Achievements:

- Designed and implemented **Angular Universal Server-Side Rendering** for SEO and social sharing
- Architected **microservices migration** using Google Cloud and Kubernetes (Java Spring, Node.js, Cloud Functions)
- Built **React and Angular component libraries** from scratch with Storybook integration
- Implemented **caching layer** leveraging Redis, reducing API response times by 60%
- Designed **Apache Kafka messaging layer** for event-driven microservices architecture
- Led **mobile DevOps automation** using Jenkins, Fastlane, achieving 90% deployment automation

Technologies: Angular 9, React, TypeScript, Java Spring Boot, Kubernetes, GCP, Kafka, Redis, Hybris, AEM, iOS (Swift), Android (Kotlin)

Senior Manager, Enterprise Applications

TD Bank

Oct 2014 - Jun 2015

- Led team of **12+ developers** delivering BI projects for banking operations
- Designed and implemented **Banking Services Data Mart** for OMEGA-PEGA project (PowerDesigner, SSIS, SSRS, SSAS)
- Developed SharePoint and ASP.NET applications for internal reporting solutions

Technologies: SQL Server, SSIS/SSAS/SSRS, MDX, .NET 4.5, ASP.NET MVC4, SharePoint, TIBCO, Cognos

Solution Architect

CAA South Central Ontario

Apr 2012 - Nov 2013

Thornhill, Canada

- Designed **complex Data Warehouse projects** in Finance, Insurance, and Travel industries
- Implemented **ESB web projects** using latest SOA and TOGAF patterns (WCF, Web Services, .NET 4.5)
- Led **SQL Server Infrastructure** design and upgrades (clustering, performance optimization)
- Architected **Enterprise Data Dictionary** solution for data warehouse governance

Technologies: SQL Server 2012, SSIS/SSAS/SSRS, QlikView, C#, .NET 4.5, WCF, TOGAF, SharePoint, Windows Azure

Availability & Contract Details

- **Available for contract work:** 10-40 hours/week
- **Remote-only:** 100% remote (Aurora, Ontario, Canada)
- **Rate:** \$80-\$120/hr depending on project scope and duration
- **Start:** Immediate availability (1-2 week notice for onboarding)
- **Contract Types:** Hourly, monthly retainer, project-based
- **Preferred Platforms:** Direct contracts, Toptal, Braintrust, Gun.io

Ideal Projects:

- Production LLM system implementation (RAG, AI agents, agentic workflows)
- AI/ML feature development for existing products
- Full-stack AI application development (Python backend + React frontend)
- Technical architecture and system design for AI-first companies
- Data engineering and ML pipeline development
- Microservices architecture and distributed systems

Education and Professional Development

Personal AI Products and Services

- <https://fasteroutcomes.com> - AI Legal Tech Platform (2024)
- <https://counta.ai> - AI Accounting Automation (2024)
- <https://frwr.ai> - AI Product (2023)
- <https://visionzlab.com> - AI Development Studio (updated 2023)
- <https://maktub.app> - SaaS Application (2018)

Certifications

- **Certified Data Scientist**, Johns Hopkins University, 2015
- **PMP Certified** (Project Management Professional), PMI, 2013
- **TOGAF 9.1 Certified**, The Open Group Architecture Framework, 2013
- **ITIL Foundation**, Mind Leaders, 2013
- **MCPD** - Web Application Development 2008, Microsoft, 2012
- **MCITP** - Database Developer 2008, Microsoft, 2012
- **Agile Project Management Strategy and Analysis**, 2013

- Data Warehousing Management, Mind Leaders, 2013
- OWASP Top 10, Security Compass, 2012

Academic Degrees

- **M.Sc. in Management**, AmirKabir University of Technology (Tehran Polytechnic), 2003
- **B.Sc. in Computer Software Engineering**, AmirKabir University of Technology (Tehran Polytechnic), 2001

Thought Leadership & Community

LinkedIn: 13K+ followers • AI/ML thought leader in production LLM systems, agentic coding, MCP development, and AI-powered development workflows

Medium: [@FrankGoortani](#) • Published technical articles on LangChain, RAG systems, AI Agents, Claude Code, agentic coding, MCP servers, production AI systems, and distributed architectures

GitHub: github.com/FrankGoortani • Open source contributions and AI/ML example repositories

Speaking & Teaching:

- Corporate Instructor - The Judge Group (Cloud Migration course, 2017-2019)
- College Instructor - George Brown College (Web & BI, 2014-2017)
- International Instructor - Learning Tree (Data Warehousing, 2014-2015)

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